HEALTH & SAFETY

Building for life means building to protect

Ensuring the safety, health and physical and mental integrity of all employees and interested parties involved in our activities is Bouygues Travaux Publics' priority. We are firmly committed the highest safety standards across each of our projects, aligned with our Clients and Partners

IMPROVE
OUR PERFORMANCES

To be recognized as the industry leader in Health and Safety, we continuously focus on 6 key areas :

- Our management system
- Safety in design
- Skill development
- Analysis of severe events and «HIPOs»
- Performance monitoring
- Support to our partners

**High-potential event*

SAFETY CULTURE

Our **General Management** impulse a strong Safety Culture based on:

- Strong leadership
- Proactive management of unexpected events
- A rigorous approach to our major risk

ENGAGE
ALL OUR EMPLOYEES

Safety relies on the involvement of each stakeholder, and we involve all our employees around the world in our health and safety ambition

Building for life means building to protect

DEVELOP A SAFETY CULTURE

OUR MAJOR RISKS

AND OUR ASSOCIATED life saving rules

1 Safety Culture

AT BOUYGUES TRAVAUX PUBLICS

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**High-potential event*

HEALTH & SAFETY POLICY
June 2024

Ensuring the safety, health and physical and mental integrity of all employees and interested parties involved in our activities is Bouygues Travaux Publics' ambition.
We aim to be recognized as a leader in health and safety in our industry and we are firmly committed to integrate our clients' requirements as well as to maintain the highest standards on all our projects.

DEVELOP A SAFETY CULTURE

Our Safety Culture approach is at the heart of the company. It promotes common ways of doing and thinking and has one ambition:

0 severe and fatal accidents

DEVELOP safety leadership based on trust and transparency
- In all our activities and all levels of the company
- An active participation and support across the chain of command
- The application of a just culture

1 GOAL Zero serious & fatal accidents

LEARN to safely manage unexpected events together
- STOP for any unforeseen event identified
- A methodology to safely mange the unexpected

FOCUS on major risks
- Life Saving Rules known and applied
- Lines of defence and Health & Safety Standards integrated into our practices

IMPROVE OUR PERFORMANCE

We are doing everything we can to continually improve our Health and Safety performance with:

1. **Our management system:** propose a structured and adapted working system covering all our activities to create and maintain safe working environment, in compliance with applicable regulations or even further;
2. **Safety in design:** prevent health and safety risks starting from the design phase and include ergonomics and arduousness workstation issues;
3. **Skill development:** train on our expertise to do safely and promote exchanges and feedback to transmit the good practices;
4. **Analysis of severe events and «HIPOs»:** analyze the accidentology and high-potential near-miss events in order to implement long-term corrective and preventive measures;
5. **Performance monitoring:** monitor indicators through the time and develop associated action plans;
6. **Support to our partners:** contribute to develop the safety approaches of stakeholders under our responsibility (subcontractors, suppliers and temporary employment agencies).

ENGAGE ALL OUR EMPLOYEES

Convinced that safety relies on the involvement of each stakeholder, we involve all our employees around the world in our health and safety ambition.

General Management guarantees to everyone the right to say «STOP».

Every employee can therefore stop a task as soon as they identify a dangerous situation for themselves or for others.

By promoting employee involvement and consultation, by capitalizing on their experience and know-how, we strengthen our ability to improve our health and safety performance.

Our managers are committed to:

- apply and enforce this policy;
- ensure the proper implementation of the management system and associated actions;
- set up the appropriate means, resources, training and skills to deploy this policy.

On a daily basis, each employee of the company plays an essential role in the implementation of this policy.

Bertrand BURTSCHELL
CEO of Bouygues Travaux Publics

OUR MAJORS RISKS

LIFTING | COLLISION | WORK AT HEIGHT | STABILITY | PRODUCTION EQUIPMENT | DANGEROUS ENERGY SOURCES

This policy is updated annually and whenever necessary.

BUILDING FOR LIFE

HEALTH & SAFETY

Building for life means building to protect

DEVELOP A SAFETY CULTURE

OUR MAIN AXIS

- Leadership
- Management of unforeseen events
- Focus on major risks

TO INVOLVE ALL EMPLOYEES

Each employee, in all sites and countries, is involved in the development and anchoring of BYTP safety Culture

2 The tools

Of our safety culture

- Axis 1 -

Leadership

All site personnel have a dedicated leadership sheet defining their responsibilities in ensuring everyone's safety.

- **Worker**
  Actor of his own safety and these of others
- **Management**
  Bearer of the BYTP vision in terms of health and safety and exemplarity
- **Support Functions**
  Actor of the correct application of his site and the project's safety rules
- **Supervision**
  Safety leader and ensurer of the correct application of the rules on his site
- **Engineering**
  Defines the operating procedures considering elimination/mitigation of risks
- **H&S teams**
  Advisor and facilitator for the appropriation of the BYTP health and safety approach

- Axis 2 -

Management of unforeseen events

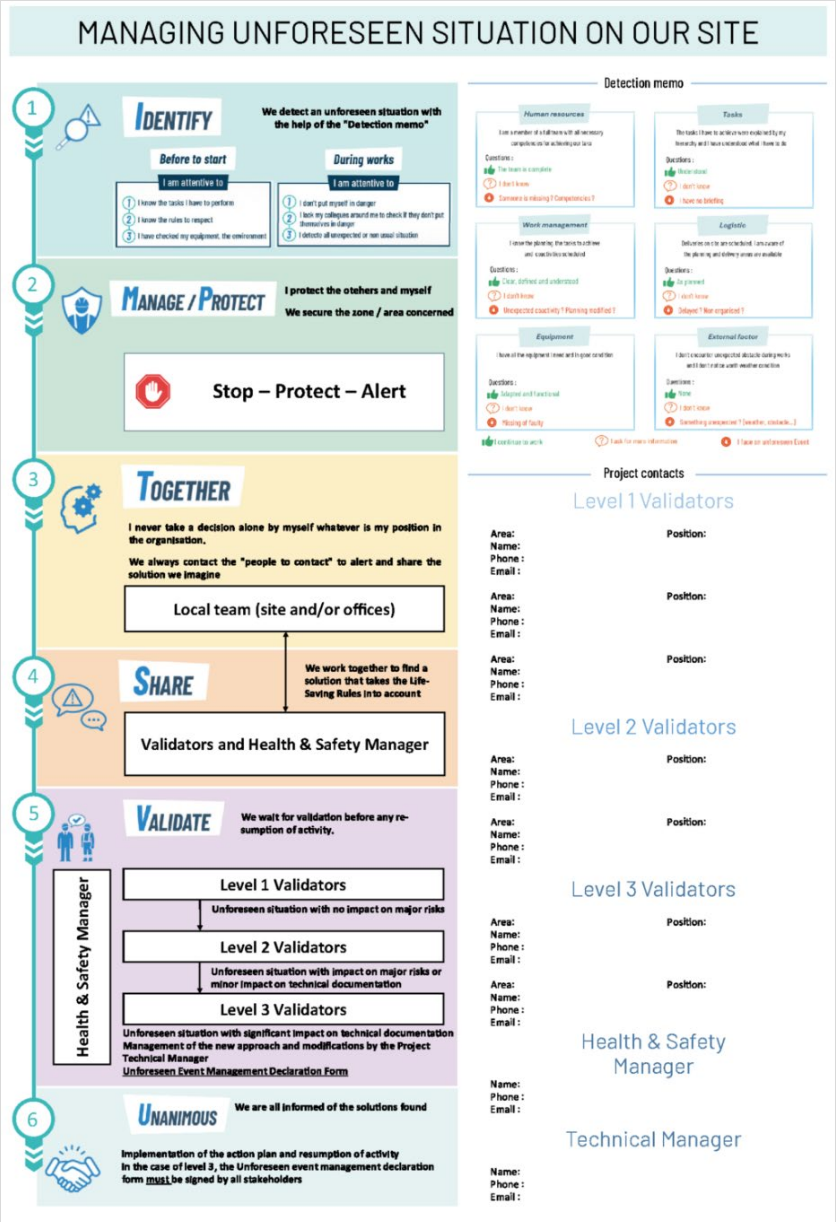
A situation is considered unforeseen when unexpected disturbing the planed activity

4 main steps to dealing with this :

- **Detection**
  Observe the working environment to detect unforeseen situations
- **Collective management**
  Define and agree together a safe solution
- **Supervision**
  Validate and supervise the application of the recovery solution
- **Partage**
  Share with the teams to avoid reoccurrence and help for future detection

This simple and adaptable approach is accessible to everyone via a single document.

It is a natural approach that is easy to implement and adaptable to different projects and local cultures.

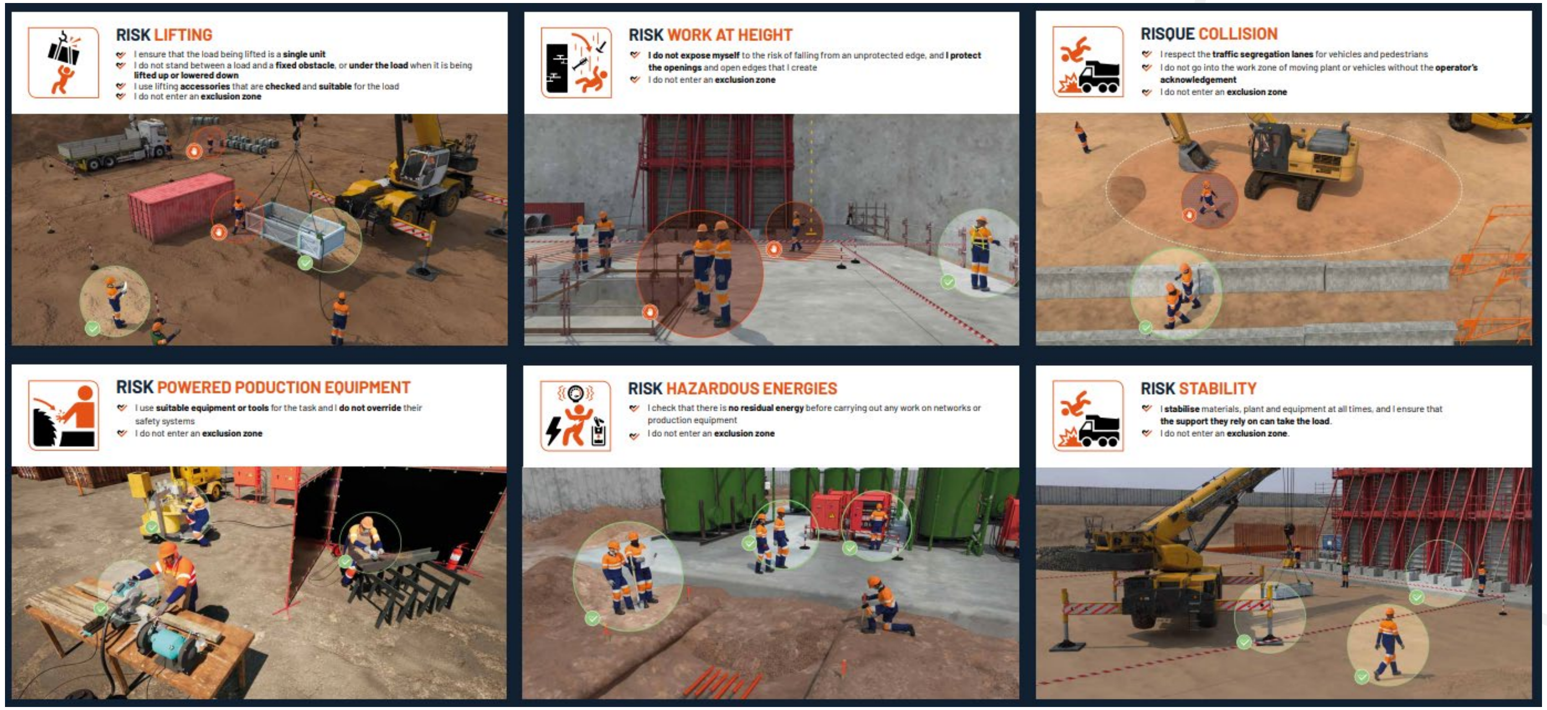


- Axis 3 -

Focus on our Major Risks

The harmonization of major risks within the Bouygues Construction group and its subsidiaries allows us a unique language on:

- our **6 risques majeurs**
- our **10 règles vitales**



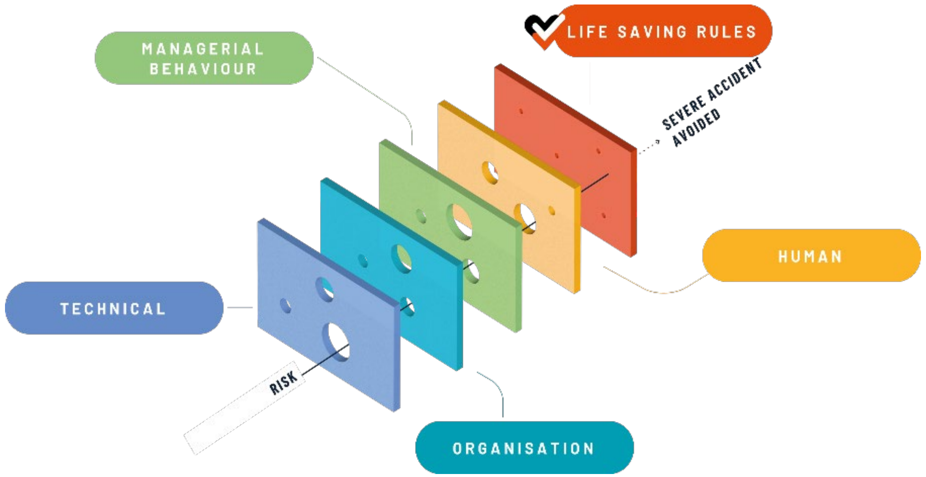
Posters **allow efficient** communication of these themes on our sites and in our offices.

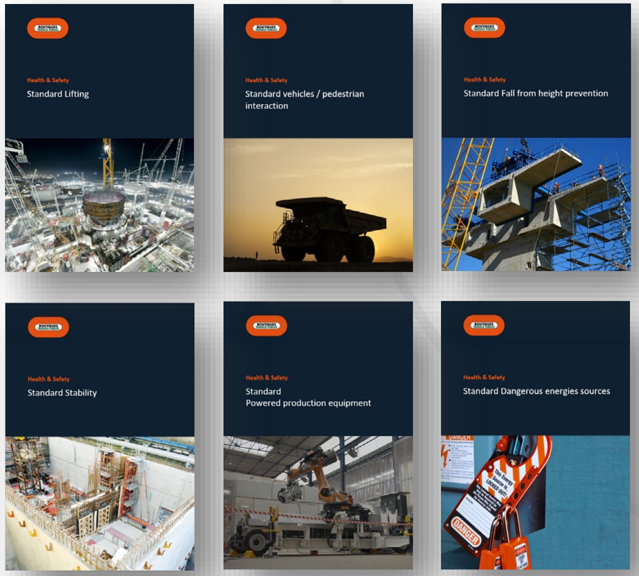
Our H&S requirements

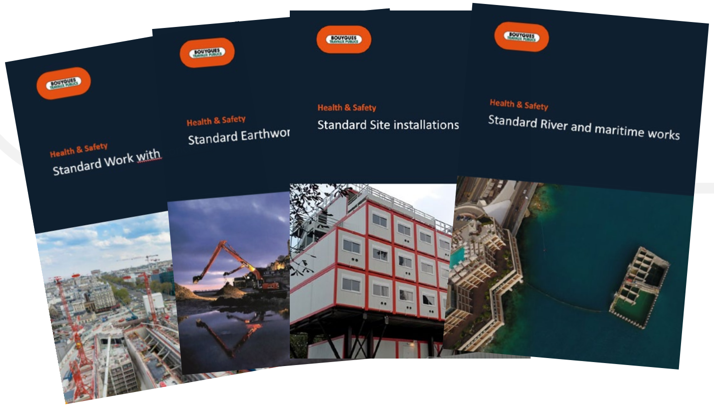
Operationnel Manual

The operational manual includes all health and safety requirements applicable to all Bouygues Travaux Publics, covering all major risks and technical activities of our core activity.

FOR	FOR	FOR
LINES OF DEFENCE	**MAJOR RISK**	**CORE ACTIVITY**







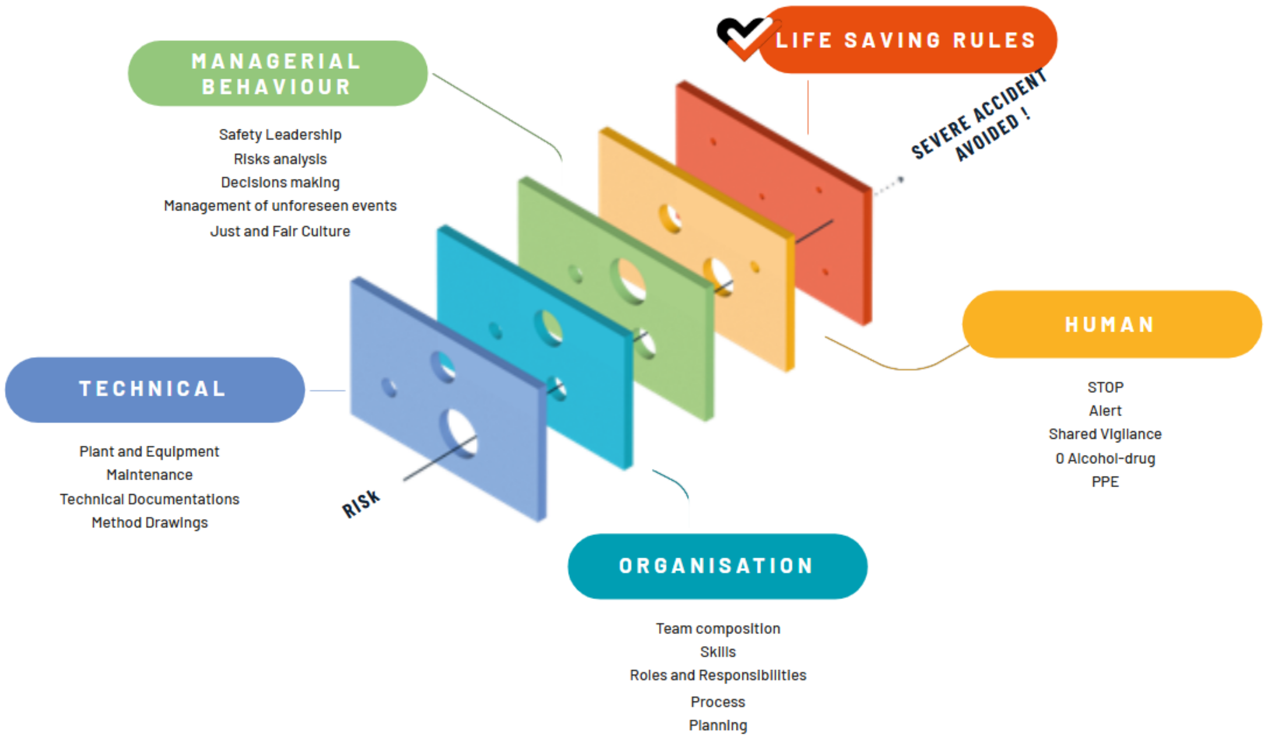
Risk management

Lines of defence

Lines of defence allow us to control our Major Risks

They are techical, organisational, managerial, human and behavioral.

Complentary and often interconnected, the prevention and protection éléments brought by these barriers are key to achieving our objective of zero severe and fatal accident.



*Our objective **« 0 accident grave »** requires us to act with humility, determination and with ever-increasing resources.*

Serious accidents can be avoided and we must do everything we can to ensure that they are.

*Let us each and every one of us implement the **technical, organisational, managerial and human lines of defense** which, supplemented by our **Life Saving Rules**, allow us to control our major risks and avoid a tragedy*

Pascal Minault
Bouygues Construction
CEO

3 Integrate Safety

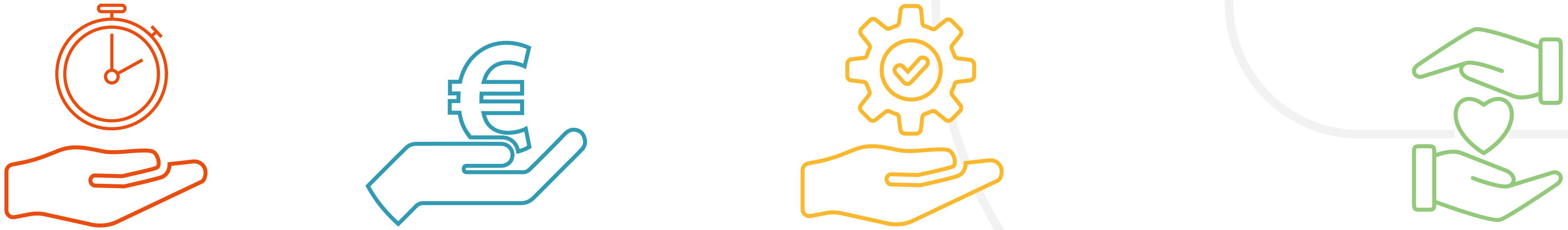
FROM THE EARLY STAGES OF PROJECTS

Integrate Safety from the early stages of projects

Safety in Design

Prevention from the start of technical studies makes it possible to eliminate, control and reduce the dangers associated with the structure throughout its life cycle, in order to ensure safer interventions.

This translates into concrete and real gains on the construction site:



In time

In cost

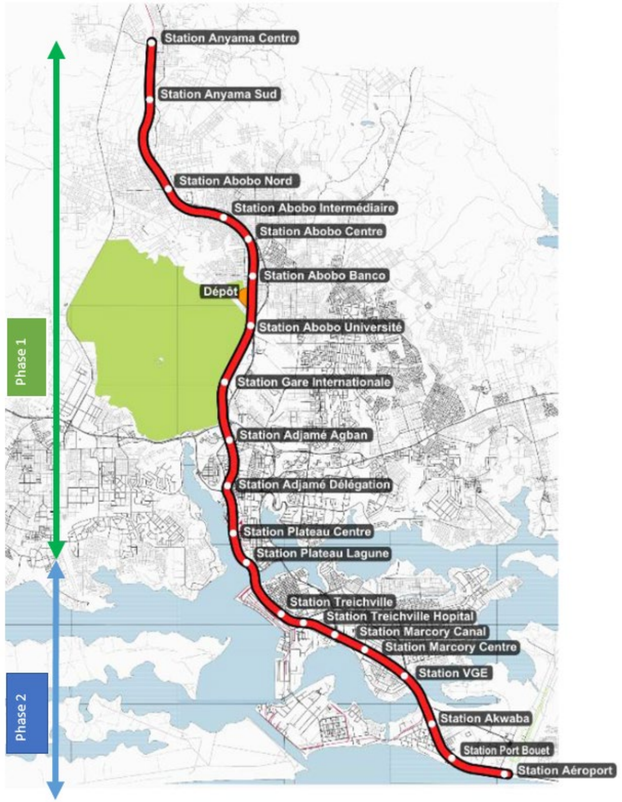
In simplification of maintenance

Safe for our teams, third parties and maintenance teams

Example : The Abidjan metro

Safety in Design at BYTP

Exemple of a high-value "SiD" project : **THE ABIDJAN METRO**



37,5 km of surface metro,
20 stations,
1 maintenance and storage center.

Key challenges:

- Users safety
- Construction methods in a congested local contexte
- Optimising engineering to aid construction by simplified operations and maintenance
- Preserving the existing railway network in the immediate vicinity of the new metro
- Respect the specifications, our requirements, as well as those of the client and the operator

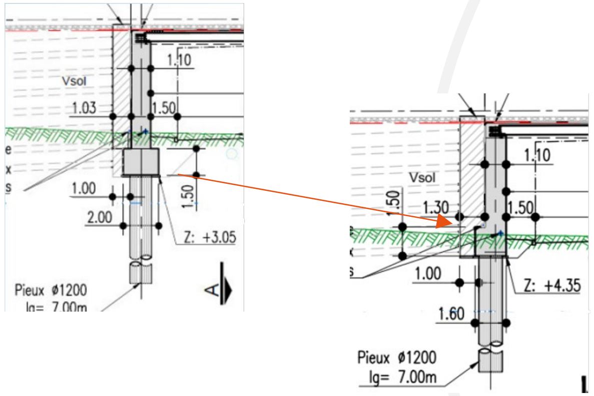
Example : The Abidjan metro

Safety in Design

Examples of solutions from the design stage of standard civil engineering structures that have a high impact on health and safety prevention:

Integral bridges solution, removal of bearings and expansion joints

Benefits
- Simplified and lower risk maintenance
- Improved robustness of the structure



Raising of footings

Benefits
- Operations can be carried out while maintaining traffic flow
- Safer working condition considering the adjacent railway
- Shallower work avoiding the need to work below the groundwater table level

Standardisation of footbridges

Benefits
- Simplification of design
- Simplification of packaging with optimised transport and installation

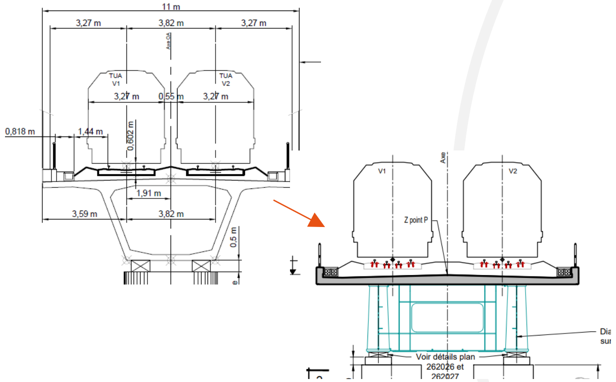
Example : The Abidjan metro

Safety in Design

Example of an impactful solutions in viaduct design over a lagoon:
Change from a prestressed box girder design to a continuous twin-girder solution

Benefits in maritime works

- Elimination of 3 ancillary structures
- Significant reduction of maritime construction resources
- Simplification of logistics and installation



Benefits in structural design

- Less technical structure without prestressing and lighter weight
- Reduction in footing heigh
- Simplified maintenance with no need for prestressed cable replacement
- Launching from the shore

Benefits from the use of prefabrication

- Prefabricated decks and troughs for footing replacing on-site casting
- Improved quality of execution
- No exposure to the aquatic environment

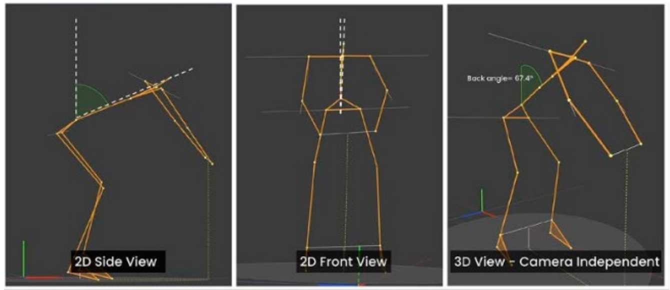
Ergonomics in Design

Integrating **ergonomics from the start** **of the design process ensures** **human-centered solutions** **that boost productivity, efficiency, and workplace health.**

Compliance with ergonomic industrial standards

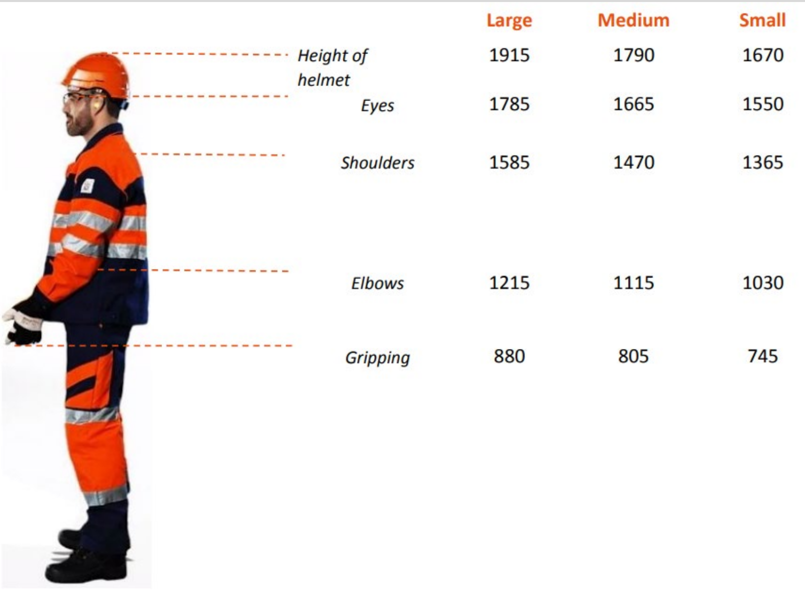
(BS EN ISO 9241 and ISO 6385: 2016)

Anticipating future changes or problems helps to reduce human and financial impacts.



Ergonomically designed systems reduce errors, downtime, and the need for rework.

They also improve the lifespan and maintainability of infrastructure by making it easier and safer to inspect and repair.



Ensuring that Anthropometric measurements are used to design for people

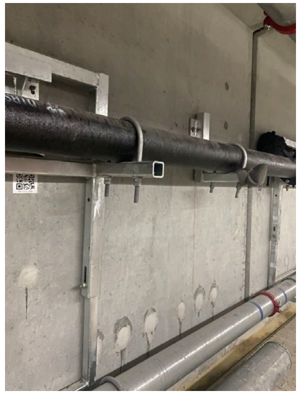
Examples

in Design at BYTP

Examples of ergonomics driven actions:

Ergonomic design of tools and equipment

Benefits
- Risk of musculoskeletal disorders reduced
- Less fatigue and accident



Modification of access and egress design to consider for human capabilities

Benefits
- Easier access for the operations maintenance
- Improve productivity

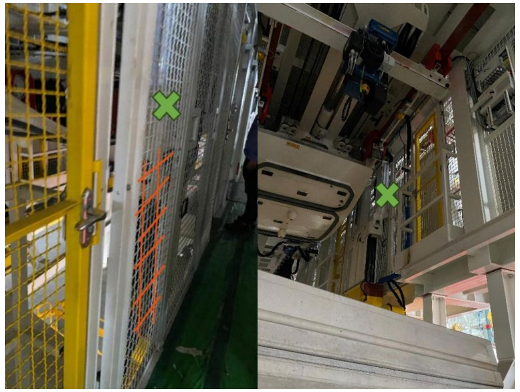
Modification of vendor material design

Benefits
- Ease the use of the equipment
- Provide feedback to vendor to improve their equipment.

Examples

Ergonomics in Design at BYTP

Examples on a TBM design



Managing better ways for timber retrieval using sustainable plastics over timber

Designing better driving cabins for comfort and displays for usage

Reducing vibration by installing anti-fatigue flooring and mats

Overall benefits

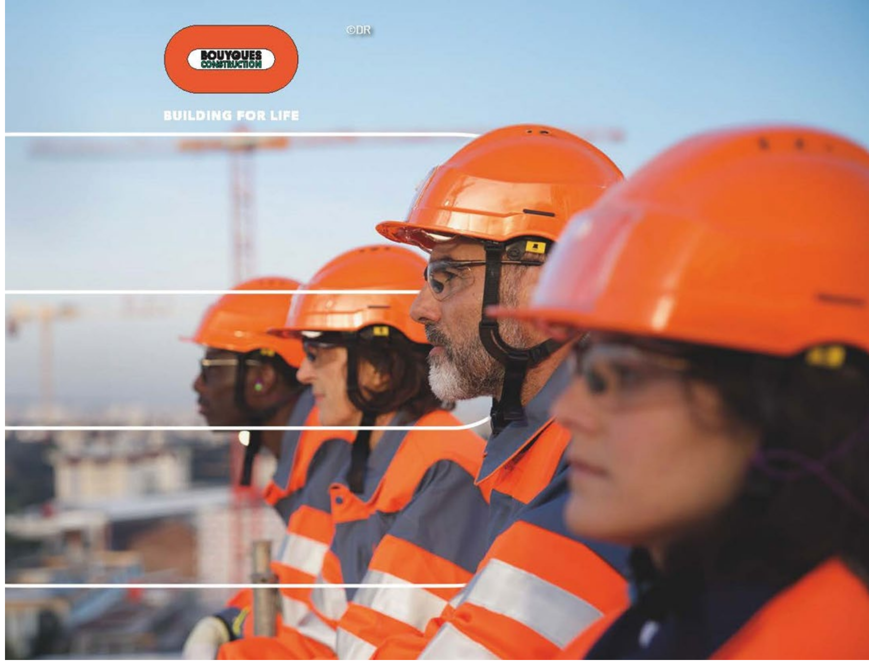
Ergonomically designed TBM improving access and egress, avoid the risk of personnel exposed to fall risks, improved working conditions, easier operations and maintenance to avoid unnecessary downtime

4 The Health and Safety Department

AT BOUYGUES TRAVAUX PUBLICS

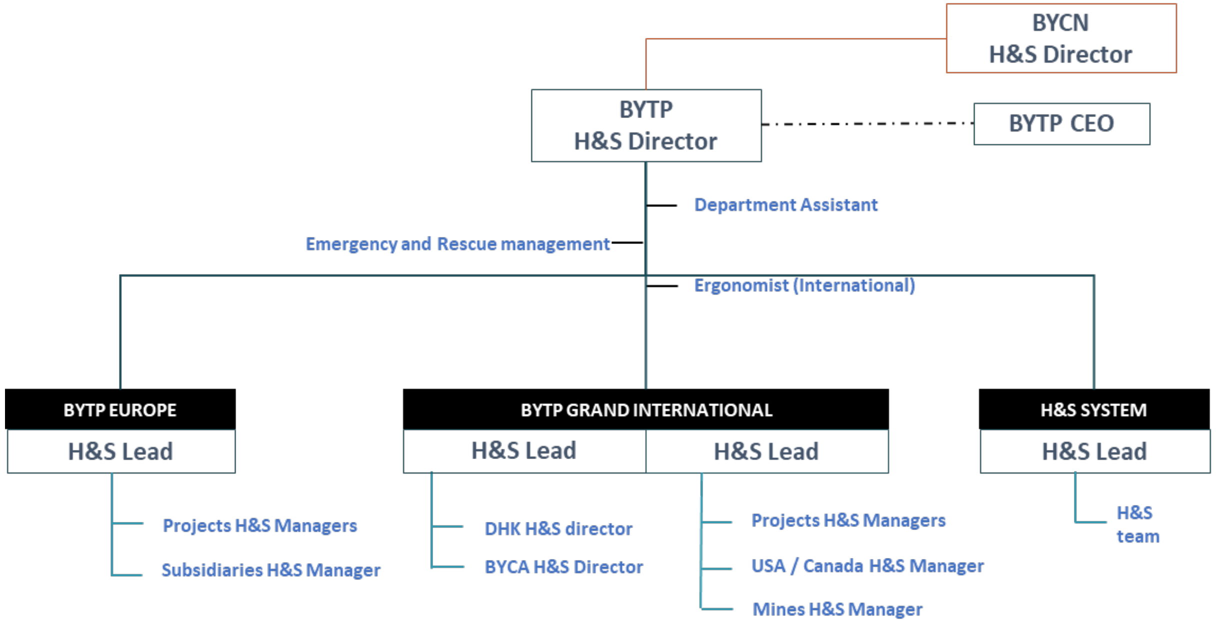
The Health and Safety department at Bouygues TP

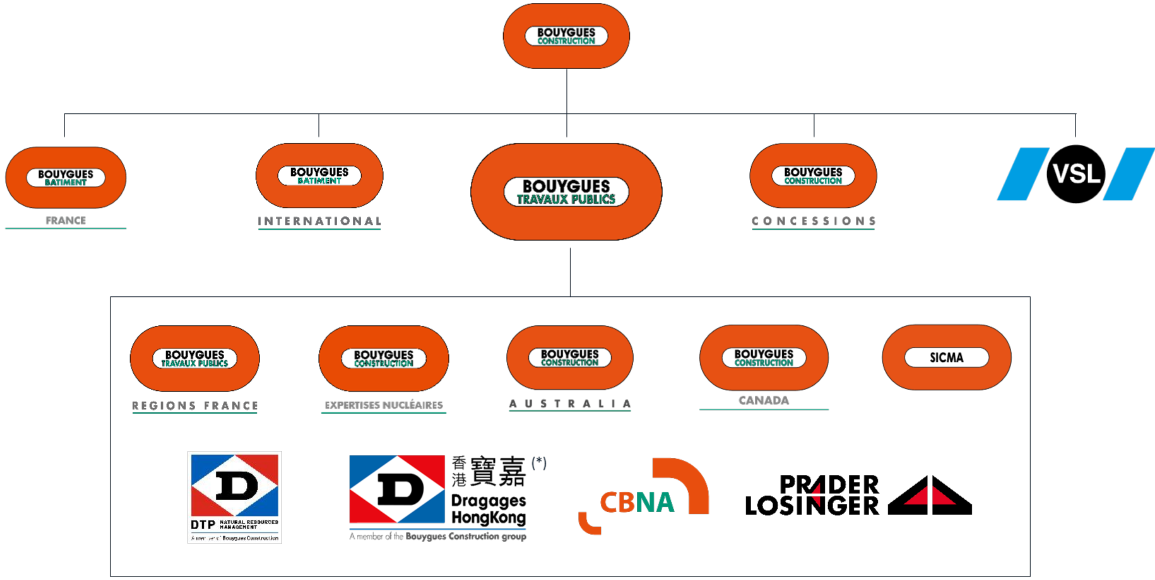
The H&S at the heart of our organization



An organisation of H&S professionals :

- Managed as a branch, independant from Operations
- Corporate/Countries/Subsidiaries functions
- Projects roles from H&S management to site supervision
- Shared expertise amongst the various entities and business of Bouygues Construction





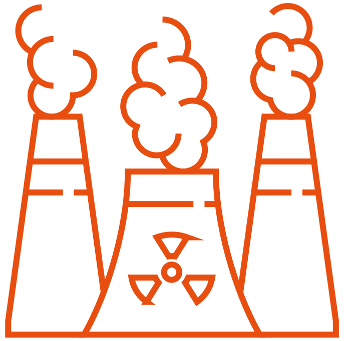
* For the Civil works activities

The Health and Safety department at Bouygues TP

The H&S at the heart of our organization

Operational H&S Hierarchy

Ergonomist



Dedicated H&S team

H&S Management system

Management of emergency situations

LOGO

5 Our interactions

AT HEART OF BOUYGUES TRAVAUX PUBLICS

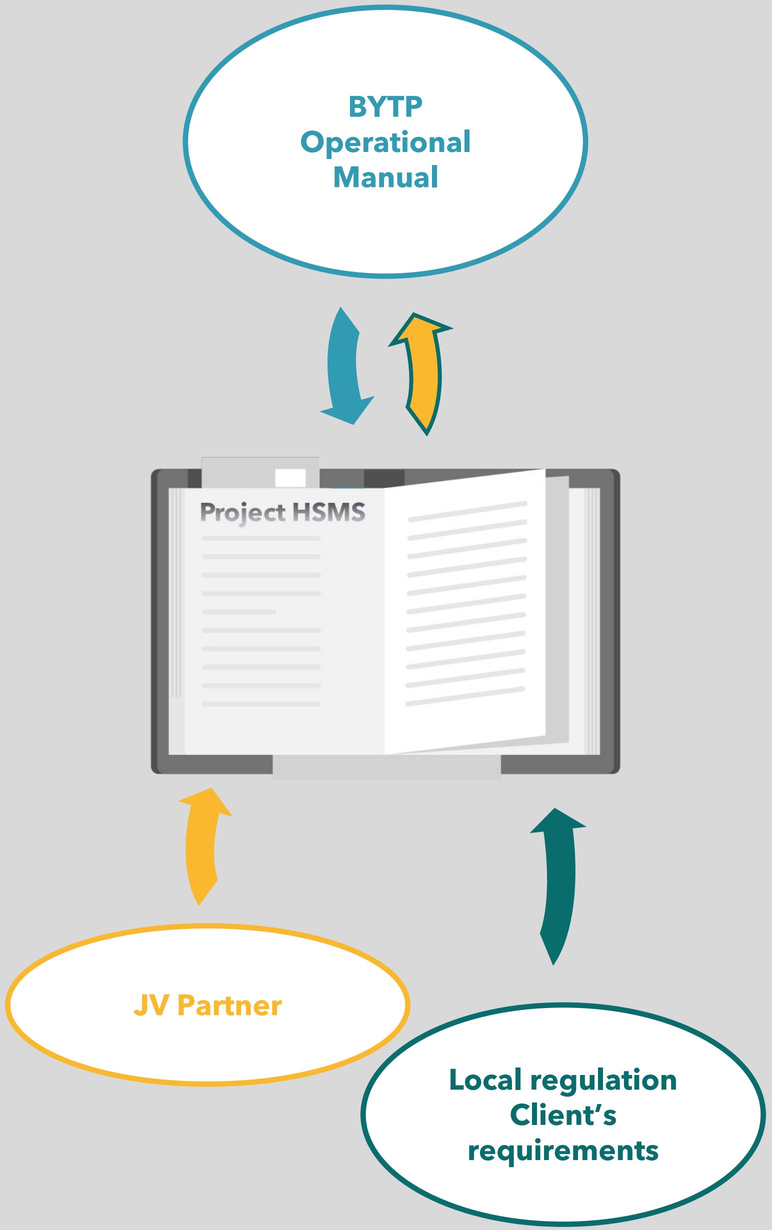
Stakeholders

THE ESSENTIALS WITH OUR PARTNER(S)

An approach implemented early in the project in order to define a common approach from the tendering phase.

Our successful collaboration with our partners is based on a fundamental agreement to prioritise health and safety and on the basis of the following requirements:

- Taking health and safety into account when making operational decisions
- Health and safety management system based on (or at least integrating) BYTP requirements (Lifesaving rules, line of defence, Company standards requirements)
- Key personnel responsible for health & safety and staffing
- Promotion of BYTP Safety Culture within the joint venture:
    - Leadership
    - Just Culture (sanction/recognition)
    - Sharing of Life Saving Rules
- Approval of the main suppliers and subcontractors
- Commitment to the Safety in Design approach



Stakeholders

MANAGEMENT AND SUPPORT OF OUR SUBCONTRACTORS

Our "0 severe or fatal accidents" Health & Safety Policy and our lines of defence apply to all stakeholders including subcontractors.

- From the tender phase: sharing our policy and contractual requirements: Health and Safety management plan, key procedures.
- Before work: kick-off meetings, site visit(s), Joint validation of H&S documents. Onboarding program before any activity begins;
- During work (including demobilisation): (self-)checks, coordination, toolbox meetings, joint analysis of significant events, etc. + participation in project life: safety days, health and safety challenges, etc.);
- At the end of the service: health and safety assessment of activities, subcontractor final evaluation

Stakeholders

AWARENESS SESSION / INFORMATION OF WORKERS

BYTP continuously takes action to inform and raise awareness among workers about the risks associated with their daily activities. Some examples of our actions:

- Health and safety onboarding process, additional health and safety modules for specific populations (e.g. temporary workers), etc.
- Training: yearly training plan including health and safety modules, awareness modules (health, addictions, on-site training , Job Safety Briefings, etc.
- Visual communication: Display of life saving rules/lines of defence, digital information screens, layout plan, emergency instructions, safety flashes
- Involvement of workers in staff representation bodies,
- Moments for discussion, sharing, and reminders of safety instructions: toolbox meetings, pre-start briefings, site visits and inspections, etc.
- Specific events: Health and safety days, safety challenges, thematic campaigns
- Dedicated health and safety staff present on site and attentive to workers.
- Lessons learned: sharing of safety flashes, best practices, and significant events that occurred on site and at BYTP

Stakeholders

AWARENESS SESSION / INFORMATION OF WORKERS

Example - Safety Week 2024

From Miami to Melbourne, via Abidjan, London, Paris, Al Ula, Hong Kong, Sydney... In other words, at all our sites, worksites and offices, Bouygues Travaux Publics staff took great pleasure in taking part in Bouygues Construction Safety Week and in reinforcing their commitment to the goal of

0 severe or fatal accidents.

During the event :

- Our executive managers took the opportunity to raise employee awareness of the line of defence that can prevent a tragedy in our business.
- Vincent AVRILLON, Safety Culture Sponsor at Bouygues Travaux Publics, gave an update on the Safety Culture launched in November 2022 and announced the harmonisation of BYTP with the BYCN life saving rules.
- In workshops, employees were able to discuss various topics related to the six major risks and ten life saving rules.
- The way our brains work and cognitive biases were presented as one of the key factors in human error.
- Finally, at the end of the event, all the participants signed the commitment poster.

6 Case studies

BYTP H&S Digitalization

Management of Health & Safety events

A unique tool for collecting all worldwide safety-related events

- Simplification
- Consistency / Reliability of data

Performances visualisation

- Safety performance directly visible for each site / project / perimeter with daily updates
- Trend analysis and focus on "hot" spots

Significant events = thorough investigations & correctives actions tracked through completion

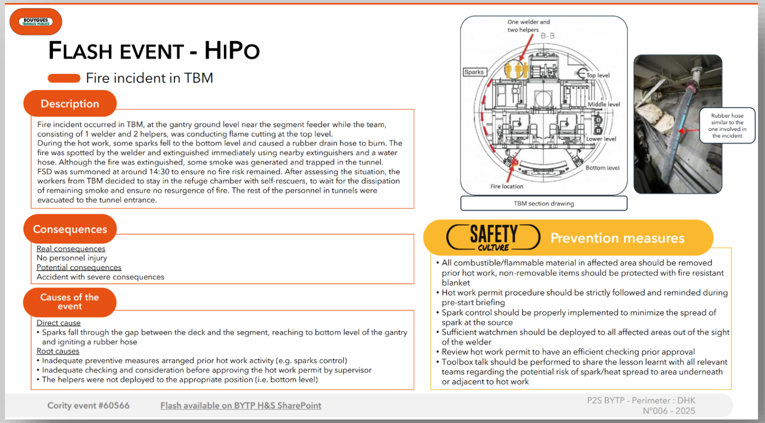
- 3 levels of validation
- All root causes analyses are captured
- Actions plan follow-up directly integrated into the tool

Lessons learned sharing

- Each H&S resource can visualize safety-related event worldwide
- Safety flashes are shared in a dedicated database by topics and severity of events, etc.

cority™

Power BI



FLASH EVENT - HIPO

Fire incident in TBM

Description

Fire incident occurred in TBM, at the gantry ground level near the segment feeder while the team, consisting of 1 welder and 2 helpers, was conducting flame cutting at the top level.
During the hot work, some sparks fell to the bottom level and caused a rubber drain hose to burn. The fire was spotted by the welder and extinguished immediately using nearby extinguishers and a water hose. Although the fire was extinguished, some smoke was generated and trapped in the tunnel.
FSD was summoned at around 14:30 to ensure no fire risk remained. After assessing the situation, the workers from TBM decided to stay in the refuge chamber with self-rescuers, to wait for the dissipation of remaining smoke and ensure no resurgence of fire. The rest of the personnel in tunnels were evacuated to the tunnel entrance.

Consequences

Real consequences
No personnel injury
Potential consequences
Accident with severe consequences

Causes of the event

Direct cause
- Sparks fall through the gap between the deck and the segment, reaching to bottom level of the gantry and igniting a rubber hose

Root causes
- Inadequate preventive measures arranged prior hot work activity (e.g. sparks control)
- Inadequate checking and consideration before approving the hot work permit by supervisor
- The helpers were not deployed to the appropriate position (i.e. bottom level)

TBM section drawing

SAFETY CULTURE **Prevention measures**

- All combustible/flammable material in affected area should be removed prior hot work, non-removable items should be protected with fire resistant blanket
- Hot work permit procedure should be strictly followed and reminded during pre-start briefing
- Spark control should be properly implemented to minimize the spread of spark at the source
- Sufficient watchmen should be deployed to all affected areas out of the sight of the welder
- Review hot work permit to have an efficient checking prior approval
- Toolbox talk should be performed to share the lesson learnt with all relevant teams regarding the potential risk of spark/heat spread to area underneath or adjacent to hot work

Cority event #60566 Flash available on BYTP H&S SharePoint

P2S BYTP - Perimeter : DHK
N°006 – 2025

Example : HS2 worksite

H&S Training center at worksite

ALIGN
HAZARD & REHEARSAL ACADEMY
aim high
Safe at heart

- **A reception area** displaying key requirements (PPE, reporting hazardous situations, etc.) and a dedicated area for each Major Risk.
- **Each area** include scenarios that allow participants to spot hazards and discuss best practices associated with Major Risks.
- The areas incorporate lessons learned from events that occurred during the project.

Satisfaction survey

What our clients say?

Surveys carried out annually with our clients by an external company

Bouygues Travaux Publics' commitment to safety is recognized.

Customer satisfaction :
We are rated 3.5/4 by our clients in terms of safety.

KEY MESSAGES

- Bouygues TP's safety management meets customer expectations...
- ...thanks to a competent and proactive safety manager and transparent communication.
- The incident with the bentonite tank allowed Bouygues TP to show the effectiveness of its crisis management and its responsiveness.
- Accidents are reported, but do not concern the customer (for the moment).
- On the contrary, the client uses them to upgrade on the other batches

---

- On the viaduct lot, transparency can (and should?) be improved.

sésame.

7 FATAL ACCIDENT

2024 LIGNE C METRO TOULOUSE LOT 7

External communication

Circumstances of the accident

General activities

- Erection of segments by crane on shoring structure and under HV lines
- Erection on simple piles
- Each segment is glued and temporarely lashed to its predecessor, then landed onto 2 hydraulic jacks
- Post-tensionning is conducted once all segments are erected
- The geometrical adjustment of the section is done once the entire
- section is erected and tensioned
- Erection from pile P77 to pile P76
- 11 segments to be erected refrenced from VSP10 - V9 to V1 –
- VSP0 (VSP = pile segment)
- Wedging of segments using hydraulic jacks
- Activity started on 26th February

Activities completed before the accident

- Erection of segments VSP10 to V1

Ongoing activity at the time of the accident

- Erection of the last segment VSP0 onto pile P76

The accident

Description

- The bridge suddenly collapsed
- The workers standing on the pathway of the shoring towers fell with the structure and landed on the ground on each side of the bridge.

Consequence

- 1 fatality and 2 workers with severe injuries

Immediate actions

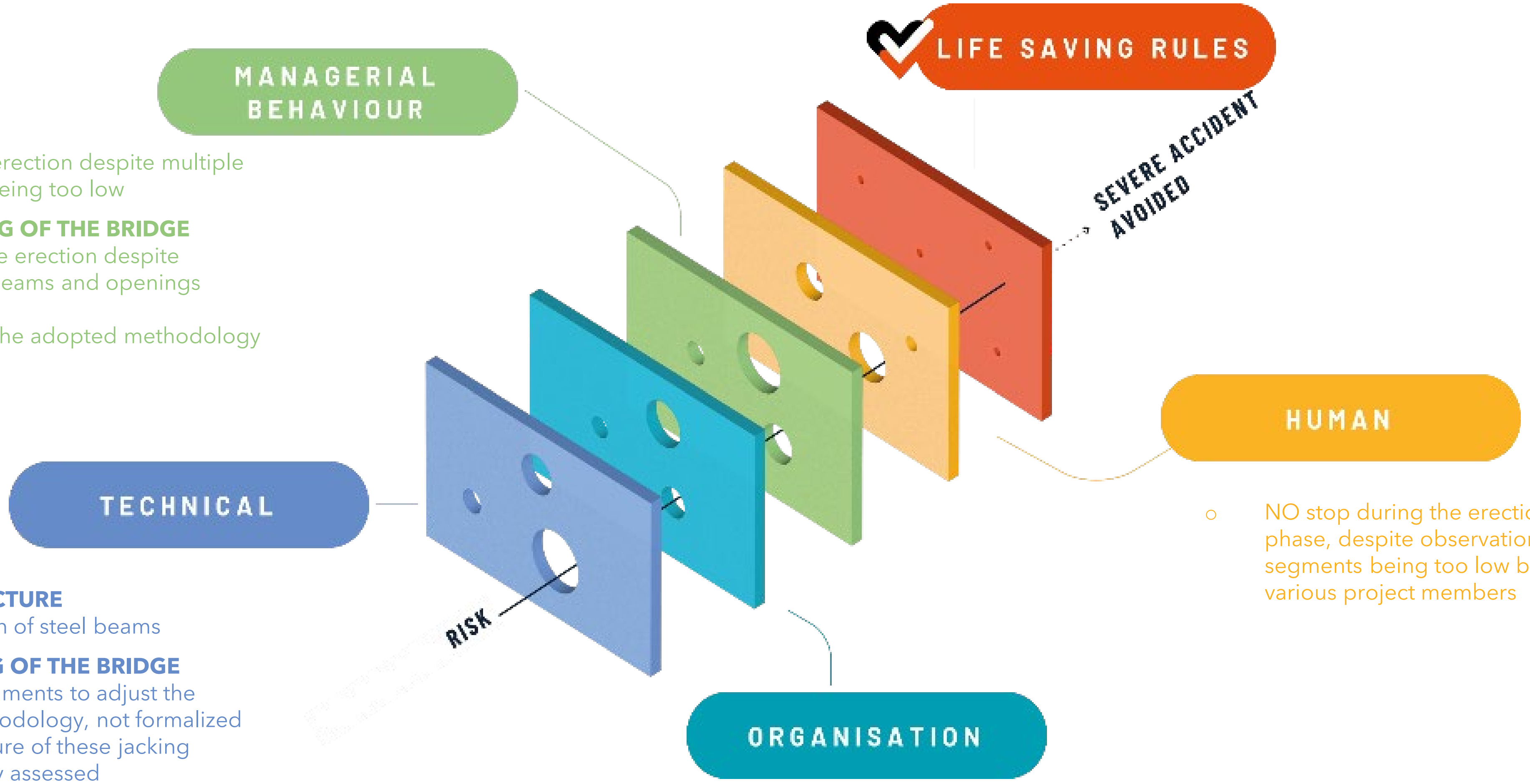
- Immediate first aid provided to the victims
- Area secured (VSP0 lowered to the ground, area barricaded and locked)
- Whole worksite stopped and accident information provided to the teams
- First interviews conducted by the police
- Crisis team triggered at BYTP Challenger
- Nomination of a dedicated investigation team

Legal Action

- A judicial investigation has been opened and is ongoing (legal appraisal, examinations, etc.)
- The investigation is not expected to be closed (nor any prosecutions initiated) before end of 2026.
- Therefore, the communication of information will remain strictly controlled.

Ligne C Metro Toulouse Lot 7 – Fatal accident

Main causes



BRIDGE GEOMETRY

- No stop during the bridge erection despite multiple observations of segments being too low

JACKING OPERATIONS / LIFTING OF THE BRIDGE

- STOP not applied during the erection despite observations of deformed beams and openings between segments
- No formal management of the adopted methodology

DESIGN OF THE SHORING STRUCTURE

- Failures in stability calculation of steel beams

JACKING OPERATIONS / LIFTING OF THE BRIDGE

- Multiple jacking/lifting of segments to adjust the geometry – unforeseen methodology, not formalized
- Impact on the shoring structure of these jacking operations were not properly assessed
- Shoring structure overloaded

- NO stop during the erection phase, despite observations of segments being too low by various project members

- Management of unforeseen situation not managed at the right level of the organisation, considering the complexity of the case
- Complex case, revision of the methodology not formalized

Ligne C Metro Toulouse Lot 7 – Accident mortel

Correctives Actions

Cause	Corrective action	Responsible person
Overload applied on the shoring structure	▪ Use of a remote-control system for monitoring efforts (measurement of hydraulic pressure in each jack or group of jacks) for the installation of segments above train tracks ▪ Specific installation procedure issued	Worksite
Jacking/lifting		
Unproper collective management of the methodology to be implemented for correcting the observed wrong geometry	▪ Set-up of limit values on the procedures and in the works execution linked to the loads on each jacks ▪ By procedure, if a limit is reached, works are stopped and engineering is consulted	Worksite
Lack of knowledge regarding the shoring structure calculation software	▪ Action with the external design office	External
	▪ Ensure internal control of calculations subcontracted to an external design office	Technical Department
No stop during the segments erections despite multiple observations of wrong geometry	▪ Use of a dedicated software to digititalize each segment and allowing the knowledge of the expected geometry of the span ▪ Set-up of limit values on the procedures and in the works execution linked to the altimetry and checked after each segment is installed	Worksite
All technical causes	▪ Lesson learn incorporated in Company standards (method statements, temporary works management procedure, etc.)	Technical Department
Management of unexpected events not carried out at the proper level of the organization	▪ Strengthening unexpected event management tools to ensure the implementation of a formal process involving the hierarchical levels required (according to the type of unexpected event)	General Management / Health & Safety Department